

The Chromatic Number of the Convex Segment Disjointness Graph[★]

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Abstract. Let P be a set of n points in general and convex position in the plane. Let D_n be the graph whose vertex set is the set of all line segments with endpoints in P , where disjoint segments are adjacent. The chromatic number of this graph was first studied by Araujo et al. [CGTA, 2005]. The previous best bounds are $\frac{3n}{4} \leq \chi(D_n) < n - \sqrt{\frac{n}{2}}$ (ignoring lower order terms). In this paper we improve the lower bound to $\chi(D_n) \geq n - \sqrt{2n}$, achieving near-tight bounds on $\chi(D_n)$.

1 Introduction

Throughout this paper, P is a set of $n > 3$ points in general and convex position in the plane. The *convex segment disjointness graph*, denoted by D_n , is the graph whose vertex set is the set of all line segments with endpoints in P , where two vertices are adjacent if the corresponding segments are disjoint. Obviously D_n does not depend on the choice of P . This graph and other related graphs, were introduced by Araujo, Dumitrescu, Hurtado, Noy and Urrutia [1], who proved the following bounds on the chromatic number of D_n :

$$2 \left\lfloor \frac{1}{3}(n+1) \right\rfloor - 1 \leq \chi(D_n) < n - \frac{1}{2} \lfloor \log n \rfloor .$$

Both bounds were improved by Dujmović and Wood [5] to

$$\frac{3}{4}(n-2) \leq \chi(D_n) < n - \sqrt{\frac{1}{2}n} - \frac{1}{2}(\ln n) + 4 .$$

In this paper we improve the lower bound, achieving near-tight bounds on $\chi(D_n)$.

Theorem 1.

$$n - \sqrt{2n + \frac{1}{4}} + \frac{1}{2} \leq \chi(D_n) < n - \sqrt{\frac{1}{2}n} - \frac{1}{2}(\ln n) + 4 .$$

[★] Dedicat al nostre amic i mestre Ferran Hurtado.

The proof of Theorem 1 is based on the observation that each colour class in a colouring of D_n is a convex thrackle. We then prove that two maximal convex thrackles must share an edge in common. From this we prove a tight upper bound on the number of edges in the union of k maximal convex thrackles. Theorem 1 quickly follows.

2 Convex Thrackles

A *convex thrackle* on P is a geometric graph with vertex set P such that every pair of edges intersect; that is, they have a common endpoint or they cross. Observe that a geometric graph H on P is a convex thrackle if and only if $E(H)$ forms an independent set in D_n . A convex thrackle is *maximal* if it is edge-maximal. As illustrated in Figure 1(a), it is well known and easily proved that every maximal convex thrackle T consists of an odd cycle $C(T)$ together with some degree 1 vertices adjacent to vertices of $C(T)$; see [2–9]. In particular, T has n edges. For each vertex v in $C(T)$, let $W_T(v)$ be the convex wedge with apex v , such that the boundary rays of $W_T(v)$ contain the neighbours of v in $C(T)$. Every degree-1 vertex u of T lies in a unique wedge and the apex of this wedge is the only neighbour of u in T .

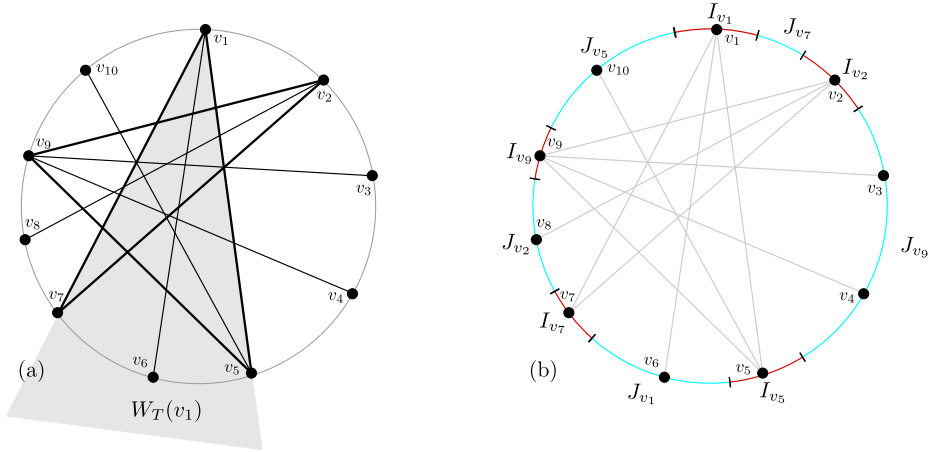


Fig. 1. (a) maximal convex thrackle with $C(T) = \{v_1, v_7, v_2, v_9, v_5\}$, (b) the intervals pairs (I_u, J_u)

3 Convex Thrackles and Free $\mathbb{Z}_2$ -Actions of S^1

In this section, we prove the key lemma (Lemma 2) that a pair of convex maximal thrackles share an edge of a particular type. We offer two proofs, one topological, one combinatorial. The topological proof first needs this technical lemma.

A $\mathbb{Z}_2$ -action on the unit circle S^1 is a homeomorphism $f : S^1 \rightarrow S^1$ such that $f(f(x)) = x$ for all $x \in S^1$. We say that f is *free* if $f(x) \neq x$ for all $x \in S^1$.

Lemma 1. *If f and g are free $\mathbb{Z}_2$ -actions of S^1 , then $f(x) = g(x)$ for some point $x \in S^1$.*

Proof. For points $x, y \in S^1$, let $\overrightarrow{xy}$ be the clockwise arc from x to y in S^1 . Let $x_0 \in S^1$. If $f(x_0) = g(x_0)$ then we are done. Now assume that $f(x_0) \neq g(x_0)$. Without loss of generality, $x_0, g(x_0), f(x_0)$ appear in this clockwise order around S^1 . Parameterise $\overrightarrow{x_0 g(x_0)}$ with a continuous injective function $p : [0, 1] \rightarrow \overrightarrow{x_0 g(x_0)}$, such that $p(0) = x_0$ and $p(1) = g(x_0)$. Assume that $g(p(t)) \neq f(p(t))$ for all $t \in [0, 1]$, otherwise we are done. Since g is free, $p(t) \neq g(p(t))$ for all $t \in [0, 1]$. Thus $g(p([0, 1])) = \overrightarrow{g(p(0))g(p(1))} = \overrightarrow{g(x_0)x_0}$. Also $f(p([0, 1])) = \overrightarrow{f(x_0)f(p(1))}$, as otherwise $g(p(t)) = f(p(t))$ for some $t \in [0, 1]$. This implies that $p(t), g(p(t)), f(p(t))$ appear in this clockwise order around S^1 . In particular, with $t = 1$, we have $f(p(1)) \in \overrightarrow{x_0 g(x_0)}$. Thus $x_0 \in \overrightarrow{f(x_0)f(p(1))}$. Hence $x_0 = f(p(t))$ for some $t \in [0, 1]$. Since f is a $\mathbb{Z}_2$ -action, $f(x_0) = p(t)$. This is a contradiction since $p(t) \in \overrightarrow{x_0 g(x_0)}$ but $f(x_0) \notin \overrightarrow{x_0 g(x_0)}$. $\square$

Assume that P lies on S^1 . Let T be a maximal convex thrackle on P . As illustrated in Figure 1(b), for each vertex u in $C(T)$, let (I_u, J_u) be a pair of closed intervals of S^1 defined as follows. Interval I_u contains u and bounded by the points of S^1 that are $1/3$ of the way towards the first points of P in the clockwise and anticlockwise direction from u . Let v and w be the neighbours of u in $C(T)$, so that v is before w in the clockwise direction from u . Let p be the endpoint of I_v in the clockwise direction from v . Let q be the endpoint of I_w in the anticlockwise direction from w . Then J_u is the interval bounded by p and q and not containing u . Define $f_T : S^1 \rightarrow S^1$ as follows. For each $v \in C(T)$, map the anticlockwise endpoint of I_v to the anticlockwise endpoint of J_v , map the clockwise endpoint of I_v to the clockwise endpoint of J_v , and extend f_T linearly for the interior points of I_v and J_v , such that $f_T(I_v) = J_v$ and $f_T(J_v) = I_v$. Since the intervals I_v and J_v are disjoint, f_T is a free $\mathbb{Z}_2$ -action of S^1 .

Lemma 2. *Let T_1 and T_2 be maximal convex thrackles on P , such that $C(T_1) \cap C(T_2) = \emptyset$. Then there is an edge in $T_1 \cap T_2$, with one endpoint in $C(T_1)$ and one endpoint in $C(T_2)$.*

Proof (Topological proof). By Lemma 1, there exists $x \in S^1$ such that $f_{T_1}(x) = y = f_{T_2}(x)$. Let $u \in C(T_1)$ and $v \in C(T_2)$ so that $x \in I_u \cup J_u$ and $x \in I_v \cup J_v$, where (I_u, J_u) and (I_v, J_v) are defined with respect to T_1 and T_2 respectively. Since $C(T_1) \cap C(T_2) = \emptyset$, we have $u \neq v$ and $I_u \cap I_v = \emptyset$. Thus $x \notin I_u \cap I_v$.

If $x \in J_u \cap J_v$ then $y \in I_u \cap I_v$, implying $u = v$. Thus $x \notin J_u \cap J_v$. Hence $x \in (I_u \cap J_v) \cup (J_u \cap I_v)$. Without loss of generality, $x \in I_u \cap J_v$. Thus $y \in J_u \cap I_v$. If $I_u \cap J_v = \{x\}$ then x is an endpoint of both I_u and J_v , implying $u \in C(T_2)$, which is a contradiction. Thus $I_u \cap J_v$ contains points other than x . It follows that $I_u \subset J_v$ and $I_v \subset J_u$. Therefore the edge uv is in both T_1 and T_2 . Moreover one endpoint of uv is in $C(T_1)$ and one endpoint is in $C(T_2)$. $\square$

Proof (Combinatorial Proof). Let H be the directed multigraph with vertex set $C(T_1) \cup C(T_2)$, where there is a *blue* arc uv in H if u is in $W_{T_1}(v)$ and there is a *red* arc uv in H if u is in $W_{T_2}(v)$. Since $C(T_1) \cap C(T_2) = \emptyset$, every vertex of H has outdegree 1. Therefore $|E(H)| = |V(H)|$ and there is a cycle Γ in the undirected multigraph underlying H . In fact, since every vertex has outdegree 1, Γ is a directed cycle. By construction, vertices in H are not incident to an incoming and an outgoing edge of the same color. Thus Γ alternates between blue and red arcs. The red edges of Γ form a matching as well as the blue edges, both of which are thrackles. However, there is only one matching thrackle on a set of points in convex position. Therefore Γ is a 2-cycle and the result follows. $\square$

4 Main Results

Theorem 2. *For every set P of n points in convex and general position, the union of k maximal convex thrackles on P has at most $kn - \binom{k}{2}$ edges.*

Proof. Let $T_1, \dots, T_k$ be a set of k maximal convex thrackles on P with m edges in total. Recall that $C(T_i)$ is the cycle in T_i .

First suppose that the cycles $C(T_1), \dots, C(T_k)$ are pairwise disjoint. By Lemma 2, distinct T_i and T_j have an edge in common, with one endpoint in $C(T_i)$ and one endpoint in $C(T_j)$. Hence distinct pairs of thrackles have distinct edges in common. Since every maximal convex thrackle has n edges and we overcount at least one edge for every pair, $m \leq kn - \binom{k}{2}$.

Now assume that some vertex is in distinct thrackles cycles. Say n_0 vertices are in no thrackle cycle. Consider a vertex v in some thrackle cycle. Let $I_v := \{i \in [1, k] : v \in C(T_i)\}$. As illustrated in Figure 2, replace v by vertices $\{v_i : i \in I_v\}$ that are placed consecutively around the circle in the spot previously occupied by v . For each $i \in T_v$ if vx and vy are the two edges in $C(T_i)$ incident to v , then replace vx and vy by v_ix and v_iy . This replacement of vertices and edges is done at all vertices simultaneously. Now, the thrackle cycles are pairwise disjoint. For each $j \in [1, k] \setminus \{i\}$, each vertex v_i is in the wedge of some vertex x in $C(T_j)$, implying the edge xv_i can be added to T_j . Hence $k - 1$ edges incident to v_i are added in this way. Moreover, these are distinct edges because the thrackle cycles are pairwise disjoint. These modified thrackles now have $n' = n_0 + \sum_v |I_v|$ vertices and $m' \geq m + \sum_v (k - 1)|I_v|$ edges. By the argument in the first case, $m' \leq kn' - \binom{k}{2}$. That is,

$$m + \sum_v (k - 1)|I_v| \leq k(n_0 + \sum_v |I_v|) - \binom{k}{2}.$$

Since $|I_v| \leq k$,

$$m \leq kn_0 + \sum_v |I_v| - \binom{k}{2} \leq kn - \binom{k}{2},$$

as desired. $\square$

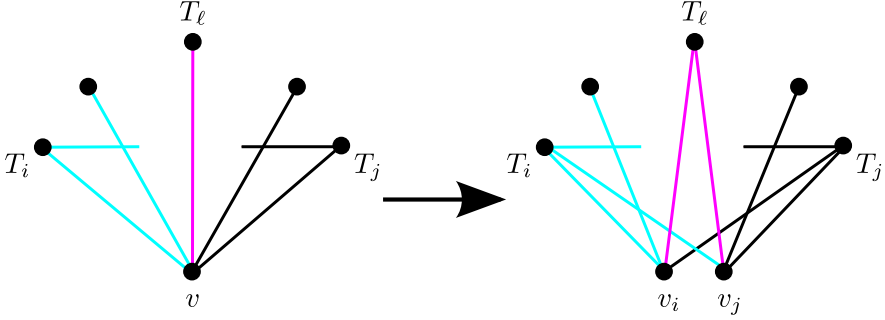


Fig. 2. Construction in the proof of Theorem 2

We now show that Theorem 2 is best possible for all $n \geq 2k$. Let S be a set of k vertices in P with no two consecutive vertices in S . If $v \in S$ and x, v, y are consecutive in this order in P , then $T_v := \{vw : w \in P \setminus \{v\}\} \cup \{xy\}$ is a maximal convex thrackle, and $\{T_v : v \in S\}$ has exactly $kn - \binom{k}{2}$ edges in total.

Proof (of Theorem 1). If $\chi(D_n) = k$ then, there are k convex thrackles whose union is the complete geometric graph on P . Possibly add edges to obtain k maximal convex thrackles with $\binom{n}{2}$ edges in total. By Theorem 2, $\binom{n}{2} \leq kn - \binom{k}{2}$. The quadratic formula implies the result. $\square$

Acknowledgements. Thanks to Javier Cano, Alfredo García, Javier Tejel and Jorge Urrutia for pointing out an error in an earlier version of this paper. R.F.-M. was supported by an Endeavour Fellowship from the Department of Education, Employment and Workplace Relations of the Australian Government. D.W. is supported by a QEII Fellowship from the Australian Research Council.

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